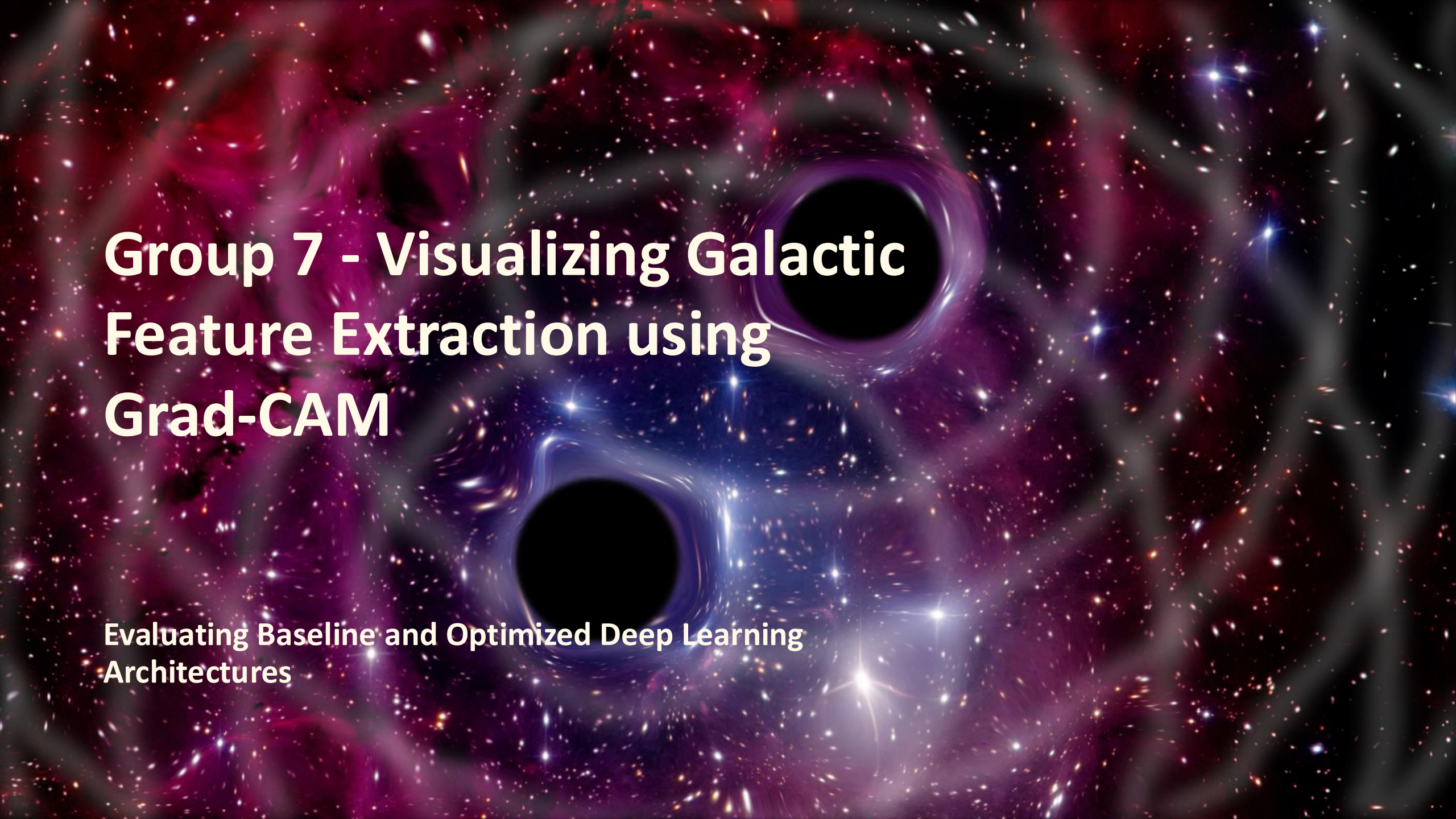


The background of the slide is a deep space image featuring a dense field of galaxies in various orientations and colors, primarily in shades of purple, magenta, and blue. Two prominent black holes are visible, each surrounded by a bright, glowing accretion disk. The overall scene is filled with numerous stars and distant galaxies, creating a sense of vastness and cosmic scale.

Group 7 - Visualizing Galactic Feature Extraction using Grad-CAM

**Evaluating Baseline and Optimized Deep Learning
Architectures**



Introduction & Motivation



Galaxy Morphology Context

Why morphology matters

Spiral, elliptical, and irregular types encode formation history, star formation activity, and interaction/merger signatures.

Scale challenge

Human labeling (experts or Galaxy Zoo) cannot keep up with LSST-scale data: billions of galaxies from terabytes per night.

Deep learning approach

Interpretability helps astronomers spot contaminants in the images, instances of distortion, and ROI which reduces error and assists in discoveries.

Problem, Bottlenecks, Goals

Scalable, interpretable galaxy morphology

The primary goal of this project is to design a scalable deep learning pipeline capable of automatically classifying galaxies into three core morphological classes: **elliptical, spiral, and irregular**. Beyond raw performance, the project emphasizes interpretability: astronomers must trust that predictions are driven by astrophysically relevant features (e.g., spiral arms or central bulges) rather than observational artifacts (e.g., background stars or noise patterns).

Objectives

- 1) Establish a strong and reliable baseline using ResNet18 to benchmark performance.
- 2) Improve the baseline using EfficientNet-B0 with transfer learning to maximize performance with limited training data.
- 3) Apply Grad-CAM to visualize model attention and validate alignment with astronomical intuition.

At first, we aimed to have models that achieved an accuracy of >70%, however, that benchmark moved with a strong ResNet18 performance to >95%.

Data and Preprocessing Pipeline

Dataset selection, preprocessing, augmentation, sampling

Dataset & Class Definition

Source & labeling

Galaxy Zoo (from SDSS imaging) provides high-quality morphology labels via consensus across thousands of independent human classifiers.

Selected classes

We used a threshold process described in Willett et al. (2013) to translate the vote distributions of the dataset into Elliptical, Spiral, and Irregular labels. The threshold chosen was >0.80 , to ensure clean definition of each class. However, that was not guaranteed.

Splits & imbalance

Train/validation/test splits prevent leakage (each image in only one split). Class imbalance—few irregulars vs. many spirals/ellipticals—must be handled to avoid majority bias.

We split the data 80/10/10 and our final set size was 16423-2053-2053.



Sample Image from training dataset

Preprocessing & Sampling

Augmentation and balanced batches

Effective preprocessing is essential for robust deep learning on astronomical images that vary in resolution, orientation, and brightness. All galaxy images were resized to 224×224 pixels to match ResNet18 and EfficientNet-B0 inputs, ensuring consistent tensor dimensions while preserving morphological detail. For EfficientNet-B0, a RandomResizedCrop was used with $\text{scale}=(0.8, 1.0)$ and $\text{ratio}=(0.9, 1.1)$.

Data augmentation (training-time)

Horizontal flips and random rotations model the fact that galaxy orientation is arbitrary and physically meaningless, pushing the network to learn intrinsic structure rather than pose-specific cues.

Weighted random sampling (class imbalance)

Underrepresented classes (especially irregular galaxies) were sampled with higher probability, producing more balanced training batches and reducing the risk of inflated accuracy from majority-class bias.

The images were also normalized based on the mean and the standard deviation of the training set. The images were already histogram equalized from the dataset, so, we did not repeat the process here.

Modeling Approach

Baseline: ResNet18

Why this baseline

Chosen for reliability in image classification and a strong balance of accuracy vs. compute, making it an ideal reference for galaxy morphology.

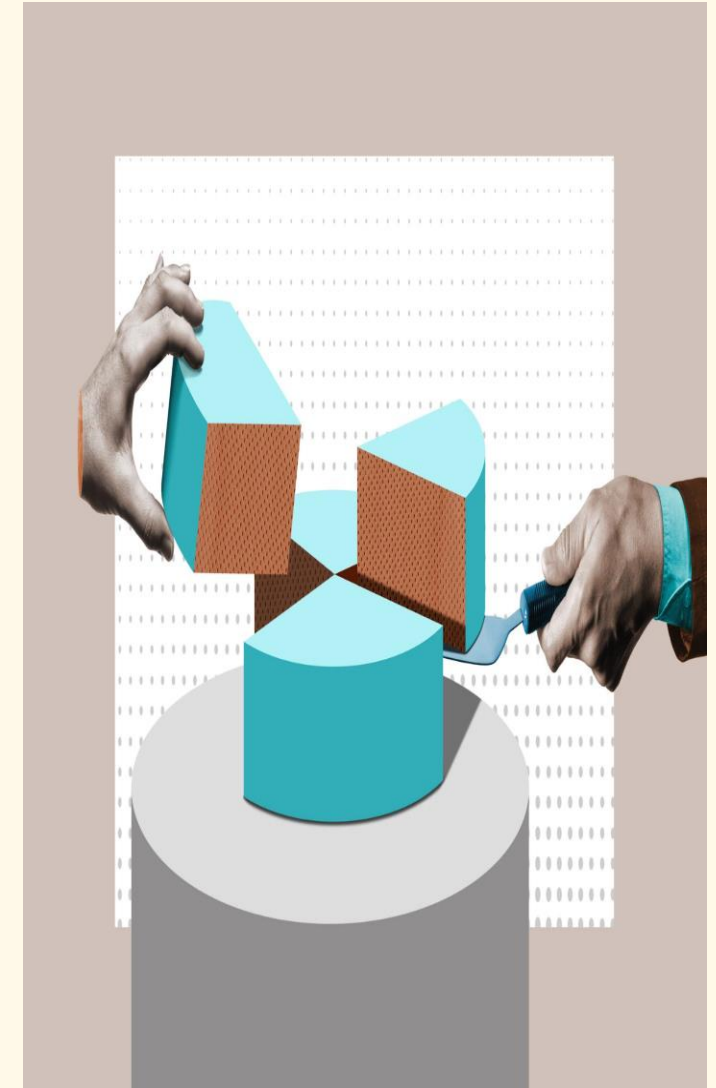
```
CFG = { 'seed': 42, 'model_name': 'resnet18', # Change this to 'efficientnet_b0'  
        later 'save_path' : '/kaggle/working/', 'model_title': 'resnet18_baseline.pth',  
        'img_size': 224, # EfficientNet-B0 also uses 224x224 'batch_size': 8, 'epochs': 8,  
        'lr': 1e-4, 'weight_decay': 1e-5, 'num_classes': 3, 'num_workers': 0 }
```

Hardware

We used Kaggle Notebooks and Kaggle's T4 GPU Accelerators for training on all the models.

Training setup

Cross-entropy loss; Adam optimizer ($lr = 1e-4$); 8 epochs to reach convergence while limiting overfitting. Serves as the benchmark for later architectures.

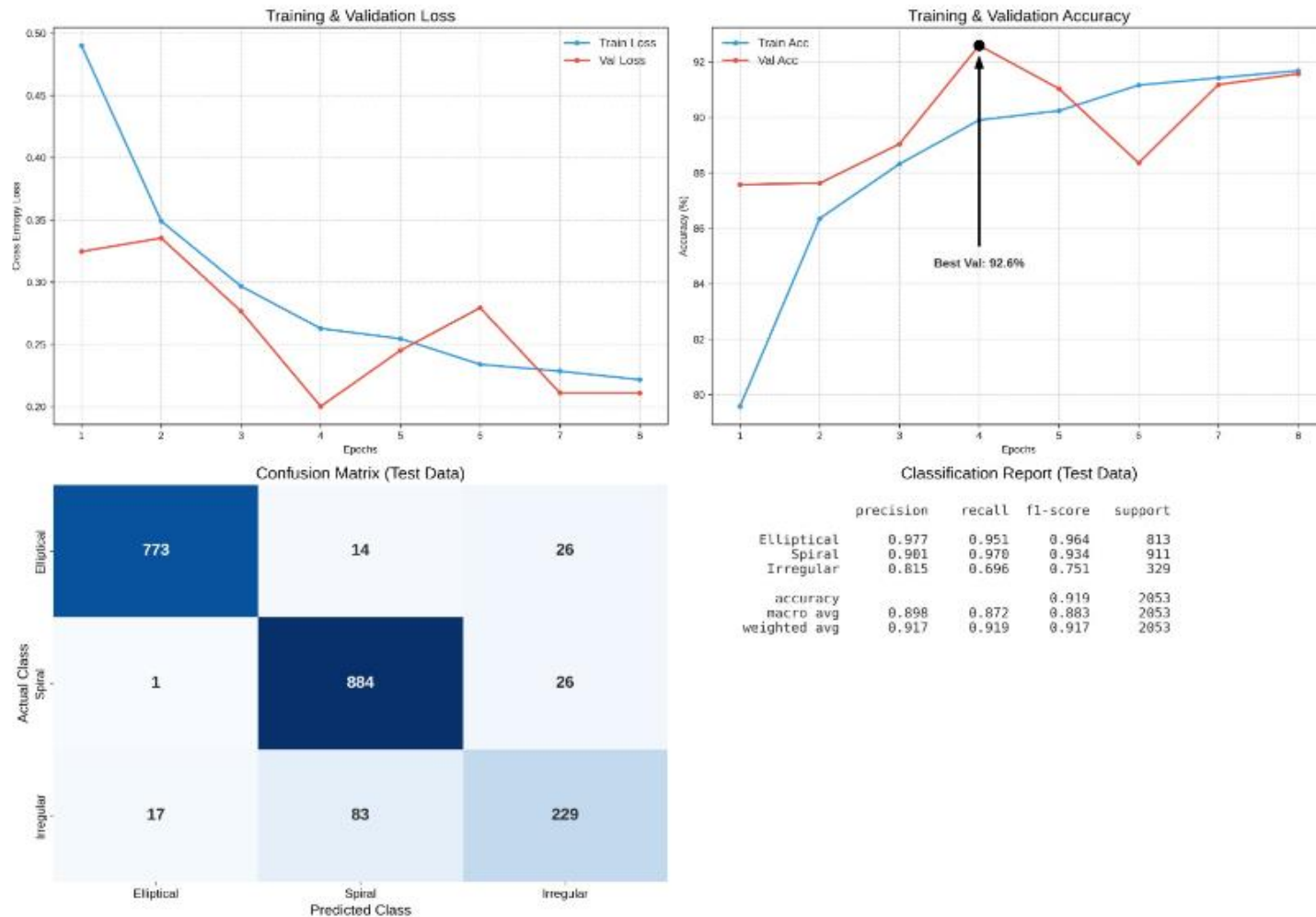


Baseline ResNet18

Model: results

- Best validation was obtained at the epoch 4/8
- Train Loss: 0.26 | Train Acc: 90%
- Val Loss: 0.3 | Val Acc: 92%
- --- **New Best Model Saved (Acc: 92%)**

Model Performance Dashboard: RESNET18



Optimized Architecture

Compound scaling efficiency

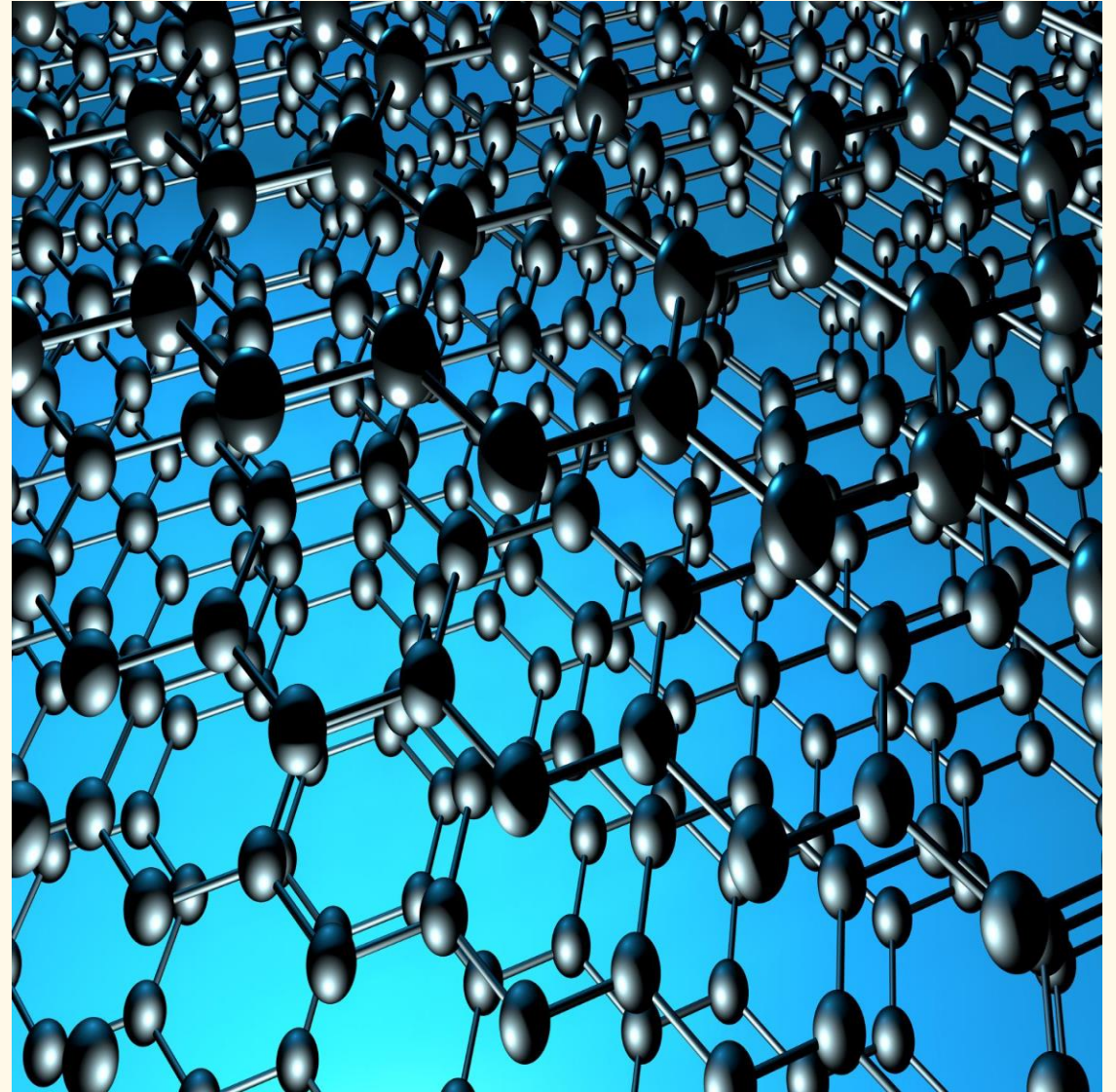
EfficientNet-B0 jointly scales depth, width, and resolution to reach strong accuracy with fewer parameters—useful when compute is limited.

Transfer learning setup

Initialized from ImageNet weights to reuse low-level features (edges, textures), then fine-tuned so higher layers adapt to galaxy structures.

Practical impact

Outperformed the ResNet18 baseline while staying compact—faster training, reduced overfitting, and better fit for survey-scale deployment.



EfficientNet model: results

Why

EfficientNet-B0 model was implemented using transfer learning and optimized hyperparameters to improve performance. Data augmentation techniques such as rotation, flipping, and cropping were applied to enhance generalization..

Parameters and training setup

Find below the chosen hyper parameters.

- 'img_size': 224,
- 'batch_size': 16, # Increased from 8 to 16 for stability
- 'epochs': 20, # Train longer for model to converge
- 'lr': 1e-4, # Good for pretrained and fine-tuning
- 'weight_decay': 1e-4, # Slightly stronger regularization
- 'num_classes': 3, # (Elliptical, Spiral, Irregular)
- 'num_workers': 0 # Keep 0 for Kaggle

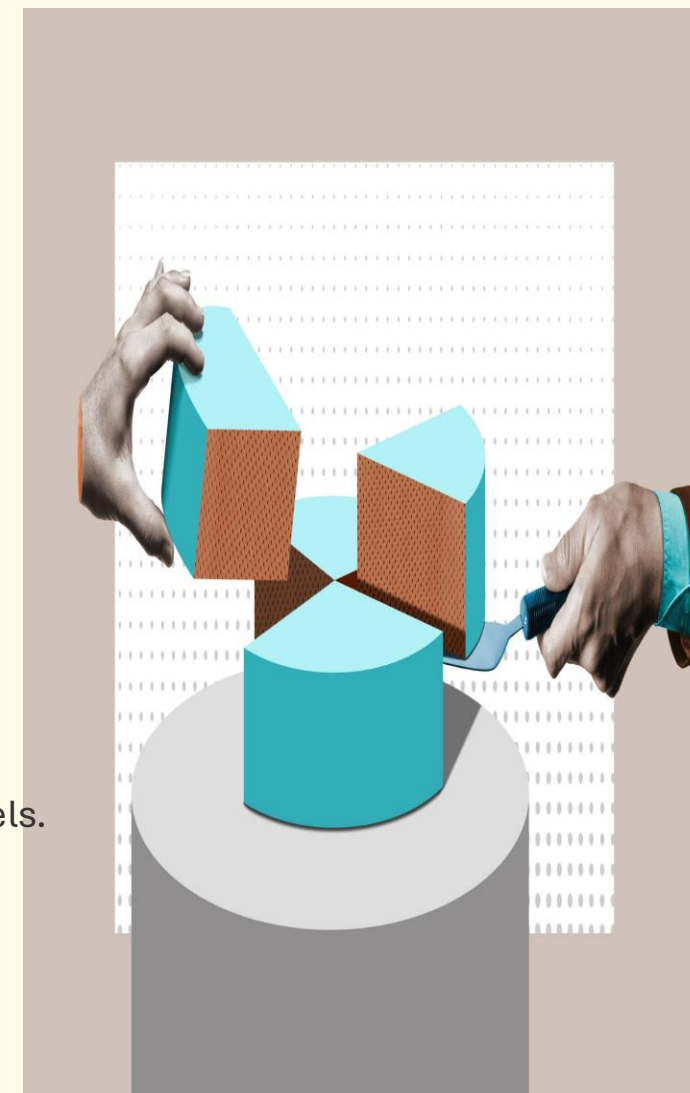
Hardware

We used Kaggle Notebooks and Kaggle's T4 GPU Accelerators for training on all the models.

Results

- Best validation was obtained at the epoch 14/20
- Epoch 14/20
- Train Loss: 0.1665 | Train Acc: 93.65
- Val Loss: 0.3082 | Val Acc: 88.50%
- --- **New Best Model Saved (Acc: 88.50%)**

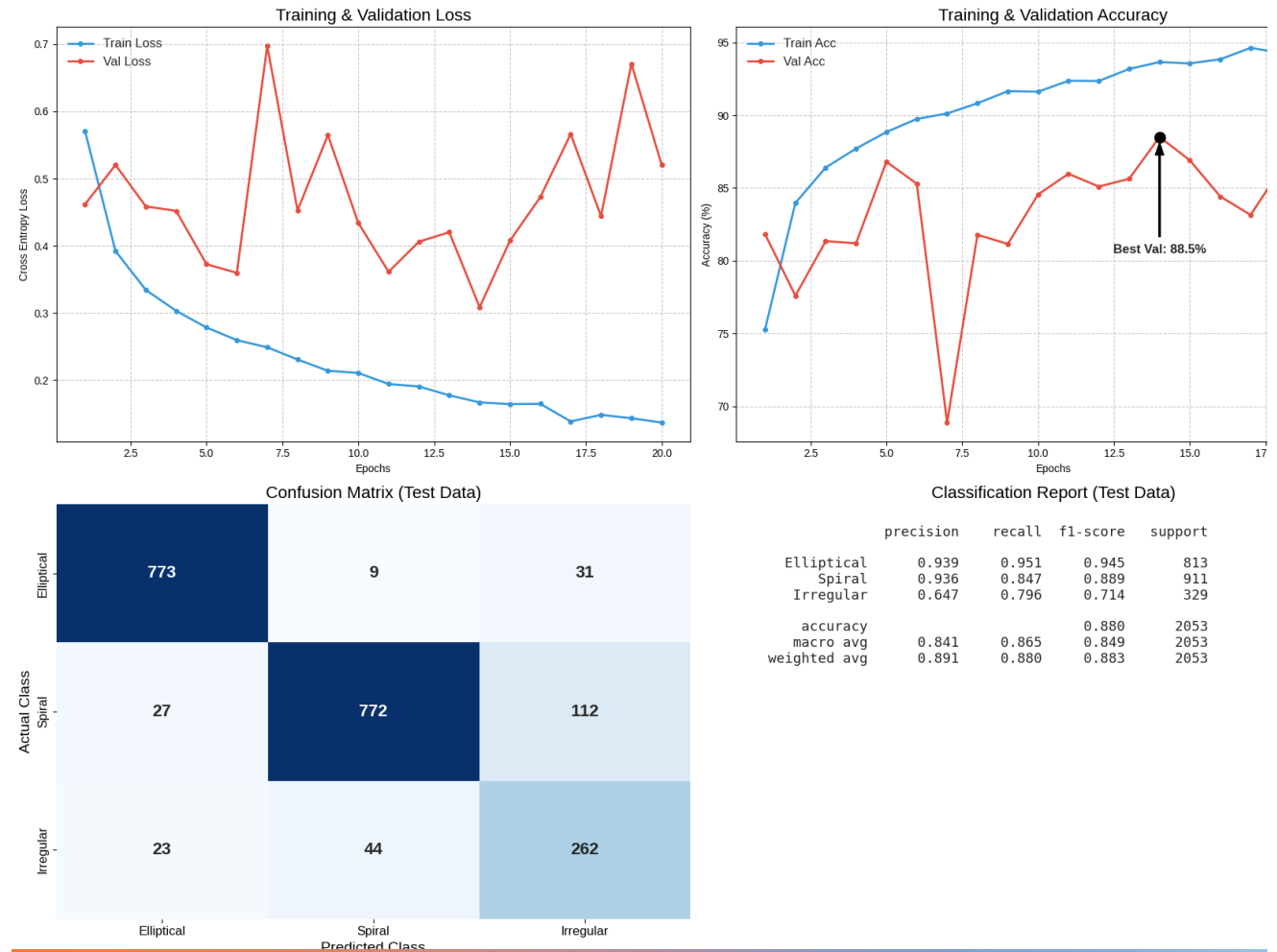
In conclusion: baseline model performed better with 92% compared to EfficientNet model 88.5%

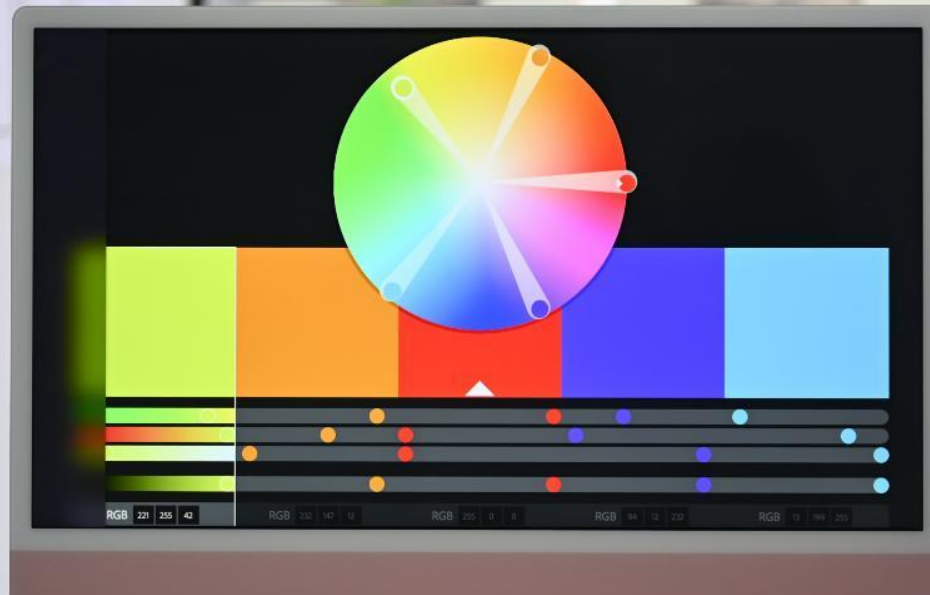


EfficientNet model

- Best validation was obtained at the epoch 14/20
- Train Loss: 0.1665 | Train Acc: 93.65
- Val Loss: 0.3082 | Val Acc: 88.50%
- --- **New Best Model Saved (Acc: 88.50%)**

Model Performance Dashboard: EFFICIENTNET_B0

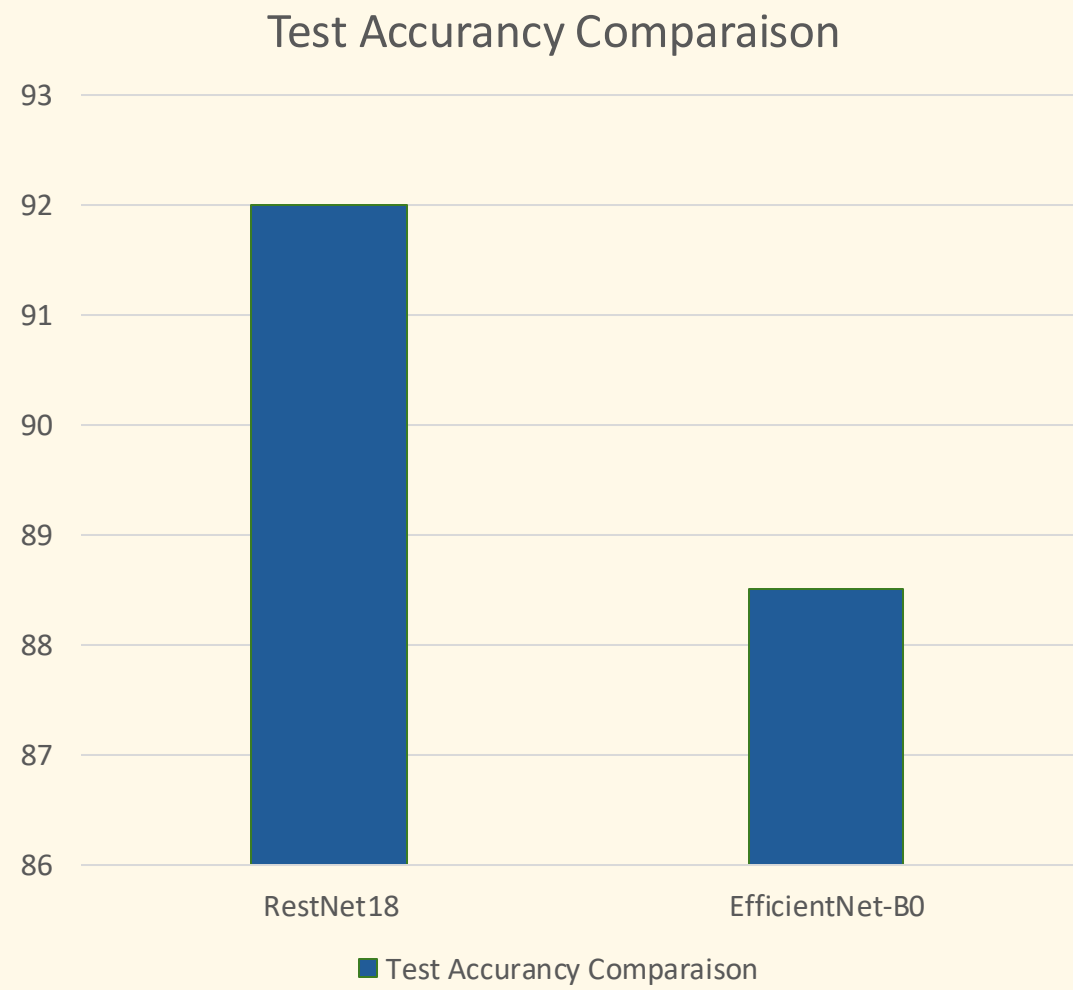




Results and Explainability

Model Performance Metrics

Model Comparison: ResNet18 vs EfficientNet-B0		
Metric	ResNet18 (Baseline)	EfficientNet-B0
Best Epoch	4 / 8	14 / 20
Train Accuracy	90%	93.65%
Train Loss	0.26	0.1665
Validation Accuracy	92%	88.50%
Validation Loss	0.3	0.3082
Training Time	Faster	Slower
Overfitting	Minimal	Observed after Epoch 14



Grad-CAM

Interpretability

Why explainability matters

Beyond strong metrics, scientific trust requires knowing what evidence drives each prediction.

How Grad-CAM works

We visualize gradients in the final convolutional layer to generate heatmaps of the most influential regions. Specifically, This done by capturing the feature maps during the forward pass and gradients during the backwards pass. With these, we can make our heatmap overlay.

What we observed

Spirals: emphasis on arms and disk. Ellipticals: focus on the dense central core; which suggests astro-physically meaningful features.



A black hole with a glowing accretion disk against a starry space background. The accretion disk is a bright, orange-yellow ring of light surrounding the dark, circular event horizon. The background is a deep black space filled with numerous small, white stars of varying brightness.

Challenges & Future Directions

Technical challenges, solutions, conclusions

Technical Challenges & Solutions

Practical fixes in a constrained workflow

Several practical challenges emerged during development, mirroring common issues in real-world deep learning workflows. In Kaggle, training triggered memory and multiprocessing errors tied to data-loading worker limits. Stability was restored by setting data loader workers to zero, accepting a performance trade-off.

Overfitting was also a concern due to the limited number of labeled irregular galaxies. This was mitigated with weight decay to discourage overly complex solutions.

The best-performing weights were saved as a checkpoint so evaluation used the most generalizable model. Overall, these outcomes emphasize that robust scientific deployment depends as much on practical engineering decisions as on theoretical model design—especially under compute and data constraints.

Translating the soft labels of the dataset into hard labels for the project risk losing the ground truth of the data. It was especially seen in the Irregular class, which had images with features of both Elliptical and Spiral classes. This is a source of bias and a likely contributor to the class' underperformance with the model.

Conclusions & Future Work

Key conclusion

Modern deep learning can effectively scale to large morphological galaxy classification with strong performance.

Model result

ResNet18 outperformed the EfficientNet-B0 baseline, achieving higher accuracy and F1 Score. Part of this is because ResNet18 is three times the size and has more parameters. However, EfficientNet-B0 does have a better recall on the minority class.

Interpretability

Grad-CAM suggests the network focuses on physically meaningful structure, supporting interpretable pipelines.

Future work

Add attention, explore ViTs, broaden labels (fine-grained or regression like merger probability), and integrate into survey pipelines.